Name:I.Harshika
Roll No:2520030607

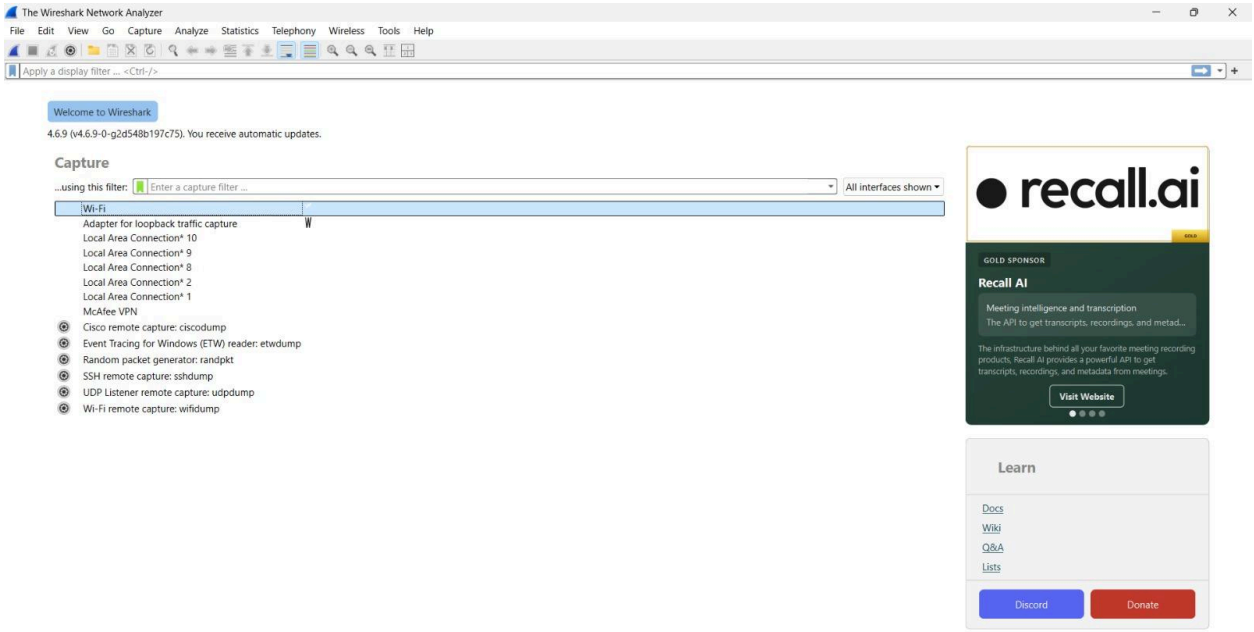
Skill - 1

Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

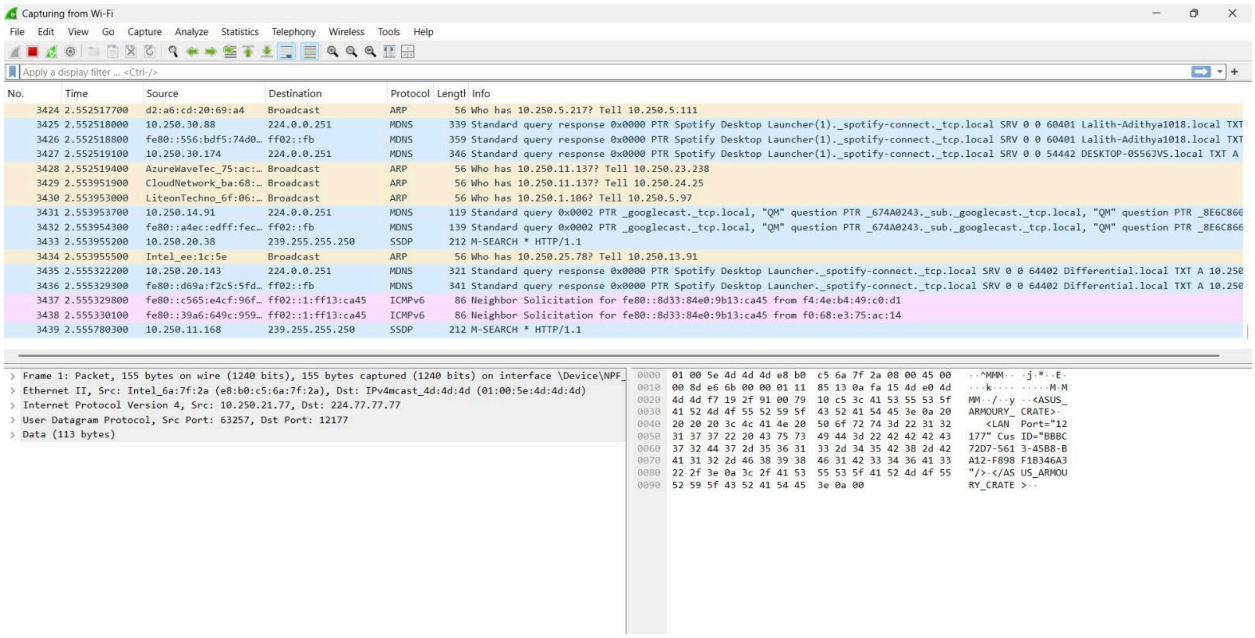
Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead. Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

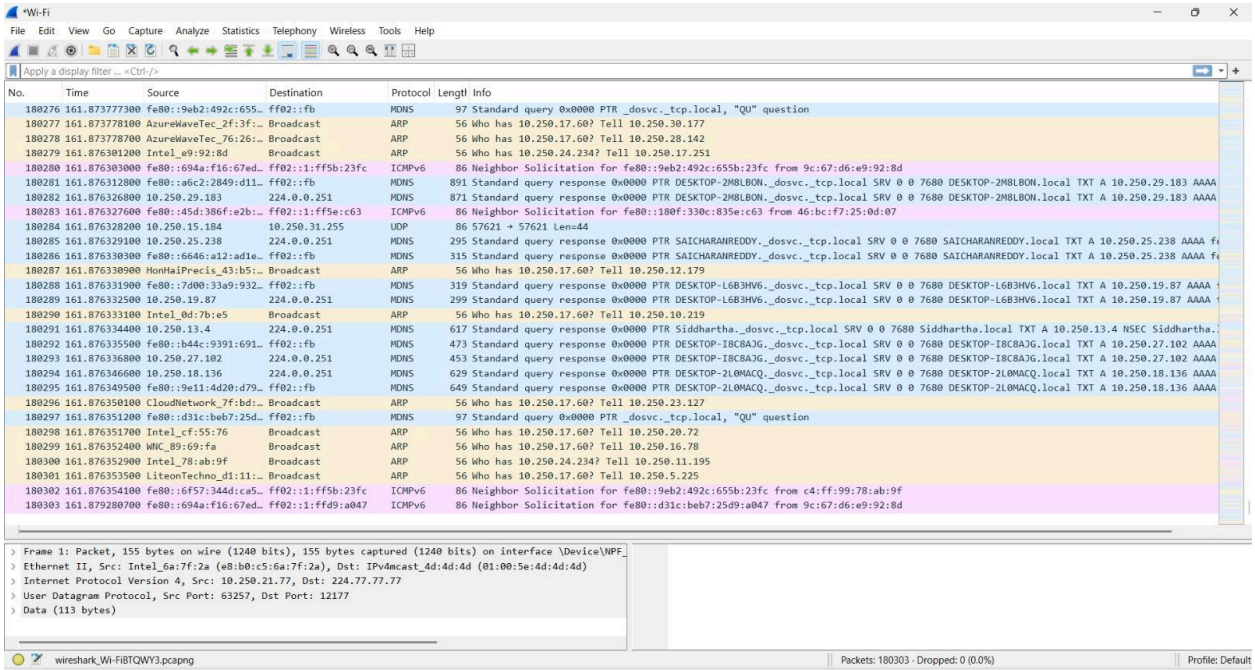
Step:1



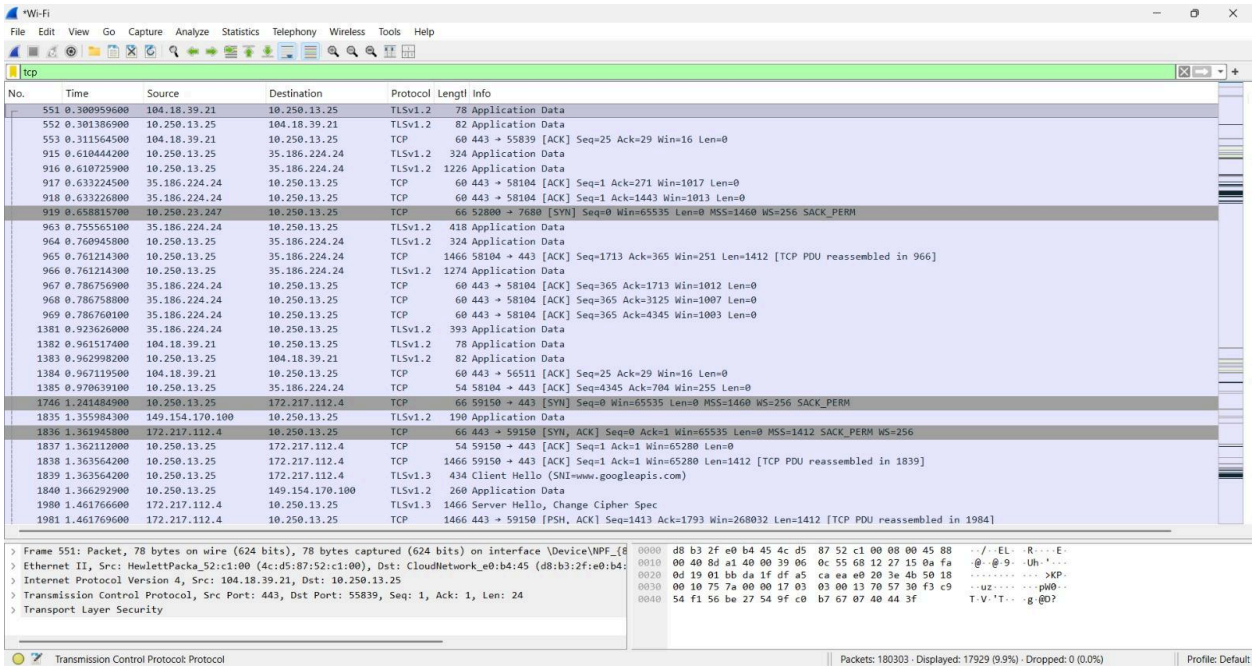
Step:2

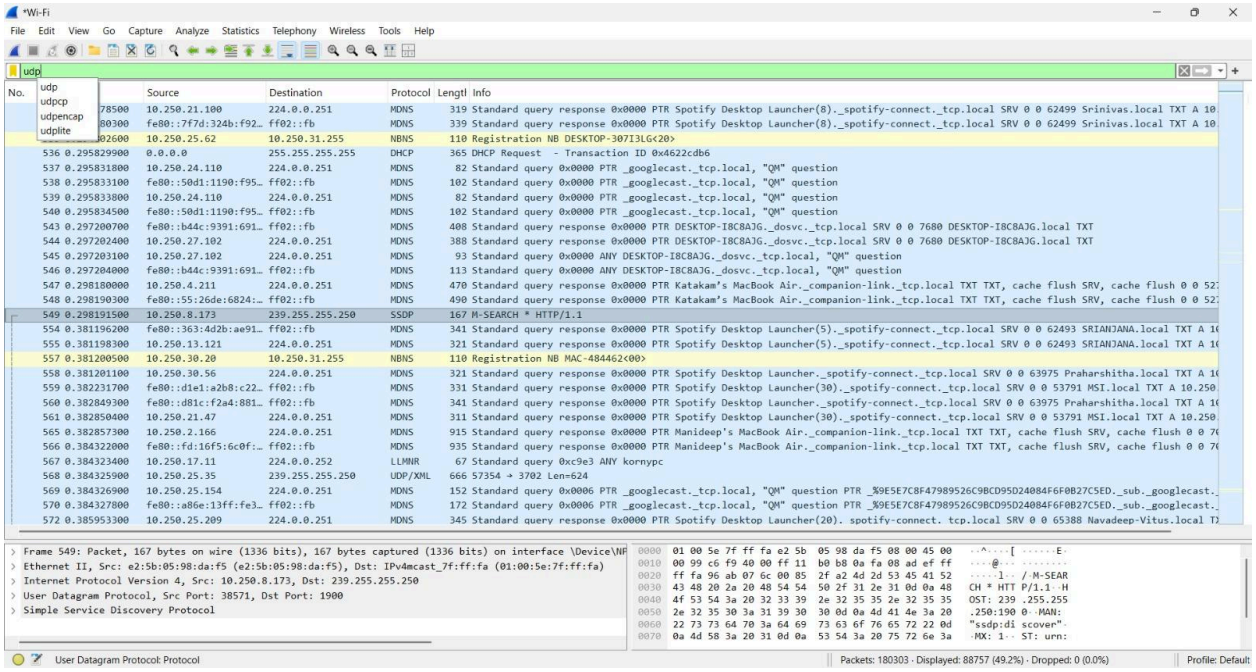

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Lengtl Info
3424 2.552517700 d2:a6:cd:20:69:a4 Broadcast ARP 56 Who has 10.250.5.217? Tell 10.250.5.111
3425 2.552518800 10.250.30.88 224.0.0.251 MDNS 339 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 60401 Lalith-Adithya1018.local TXT
3426 2.552518800 fe80::556:bdf5:74d0… ff02::fb MDNS 359 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 60401 Lalith-Adithya1018.local TXT
3427 2.552519100 10.250.30.174 224.0.0.251 MDNS 346 Standard query response 0x0000 PTR Spotify Desktop Launcher(1)._spotify-connect._tcp.local SRV 0 0 54442 DESKTOP-0S56JVS.local TXT A
3428 2.552519400 AzureWaveTec_75:ac:… Broadcast ARP 56 Who has 10.250.11.137? Tell 10.250.23.238
3429 2.553951900 CloudNetwork_ba:68:… Broadcast ARP 56 Who has 10.250.11.137? Tell 10.250.24.25
3430 2.553953000 LiteonTechno_6f:06:… Broadcast ARP 56 Who has 10.250.1.106? Tell 10.250.5.97
3431 2.553953700 10.250.14.91 224.0.0.251 MDNS 119 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question PTR _8E6C866
3432 2.553954300 fe80::a4ec:edff:fec… ff02::fb MDNS 139 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question PTR _8E6C866
3433 2.553955200 10.250.20.38 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
3434 2.553955500 Intel_ee:1c:5e Broadcast ARP 56 Who has 10.250.25.78? Tell 10.250.13.91
3435 2.555322200 10.250.20.143 224.0.0.251 MDNS 321 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 64402 Differential.local TXT A 10.250
3436 2.555329300 fe80::d69a:f2c5:5fd… ff02::fb MDNS 341 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 64402 Differential.local TXT A 10.250
3437 2.555329800 fe80::c565:e4cf:96f… ff02::1:ff13:ca45 ICMPv6 86 Neighbor Solicitation for fe80::8d33:84e0:9b13:ca45 from f4:4e:b4:49:c0:d1
3438 2.555330100 fe80::39a6:649c:959… ff02::1:ff13:ca45 ICMPv6 86 Neighbor Solicitation for fe80::8d33:84e0:9b13:ca45 from f0:68:e3:75:ac:14
3439 2.555780300 10.250.11.168 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
Frame 1: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Device\NPF
Ethernet II, Src: Intel_6a:7f:2a (e8:b0:c5:6a:7f:2a), Dst: IPv4mcast_4d:4d:4d (01:00:5e:4d:4d:4d)
Internet Protocol Version 4, Src: 10.250.21.77, Dst: 224.77.77.77
User Datagram Protocol, Src Port: 63257, Dst Port: 12177
Data (113 bytes)
0000 01 00 5e 4d 4d 4d e8 b0 c5 6a 7f 2a 08 00 45 00 ··^MMM·· ·j·*··E·
0010 00 8d e6 6b 00 00 01 11 85 13 0a fa 15 4d e0 4d ···k···· ·····M·M
0020 4d 4d f7 19 2f 91 00 79 10 c5 3c 41 53 55 53 5f MM··/··y ··<ASUS_
0030 41 52 4d 4f 55 52 59 5f 43 52 41 54 45 3e 0a 20 ARMOURY_ CRATE>·
0040 20 20 20 3c 4c 41 4e 20 50 6f 72 74 3d 22 31 32 <LAN Port="12
0050 31 37 37 22 20 43 75 73 49 44 3d 22 42 42 42 43 177" Cus ID="BBBC
0060 37 32 44 37 2d 35 36 31 33 2d 34 35 42 38 2d 42 72D7-561 3-45B8-B
0070 41 31 32 2d 46 38 39 38 46 31 42 33 34 36 41 33 A12-F898 F1B346A3
0080 22 2f 3e 0a 3c 2f 41 53 55 53 5f 41 52 4d 4f 55 "/>·</AS US_ARMOU
0090 52 59 5f 43 52 41 54 45 3e 0a 00 RY_CRATE >··

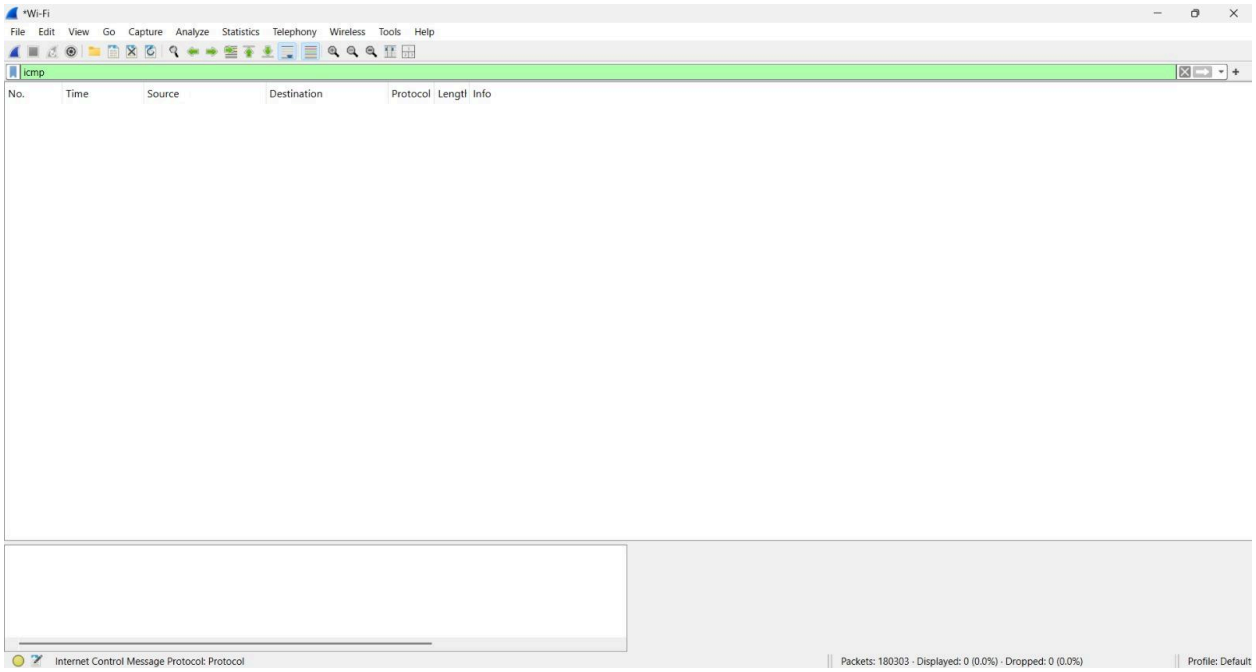
Step:3

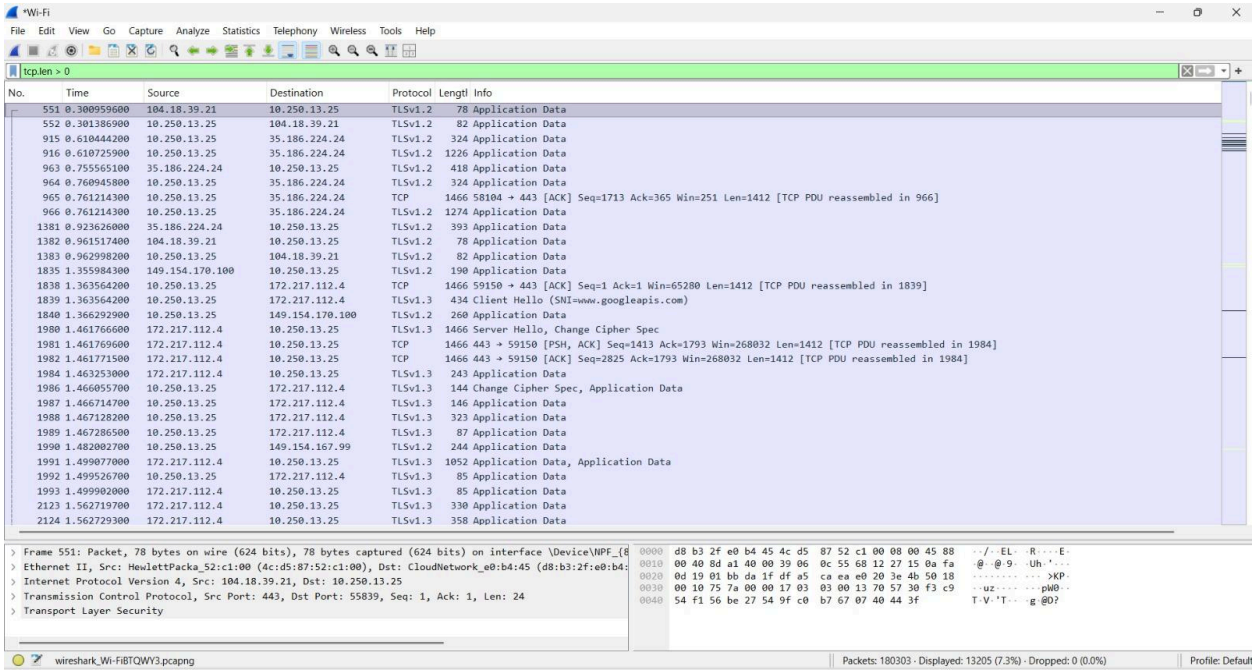

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Lengtl Info
180276 161.873777300 fe80::9eb2:492c:655… ff02::fb MDNS 97 Standard query 0x0000 PTR _dosvc._tcp.local, "QU" question
180277 161.873778100 AzureWaveTec_2f:3f:… Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.30.177
180278 161.873778700 AzureWaveTec_76:26:… Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.28.142
180279 161.876301200 Intel_e9:92:8d Broadcast ARP 56 Who has 10.250.24.234? Tell 10.250.17.251
180280 161.876303000 fe80::694a:f16:67ed… ff02::1:ff5b:23fc ICMPv6 86 Neighbor Solicitation for fe80::9eb2:492c:655b:23fc from 9c:67:d6:e9:92:8d
180281 161.876312800 fe80::a6c2:2849:d11… ff02::fb MDNS 891 Standard query response 0x0000 PTR DESKTOP-2M8LBON._dosvc._tcp.local SRV 0 0 7680 DESKTOP-2M8LBON.local TXT A 10.250.29.183 AAAA
180282 161.876326800 10.250.29.183 224.0.0.251 MDNS 871 Standard query response 0x0000 PTR DESKTOP-2M8LBON._dosvc._tcp.local SRV 0 0 7680 DESKTOP-2M8LBON.local TXT A 10.250.29.183 AAAA
180283 161.876327600 fe80::45d:386f:e2b:… ff02::1:ff5e:c63 ICMPv6 86 Neighbor Solicitation for fe80::180f:330c:835e:c63 from 46:bc:f7:25:0d:07
180284 161.876328200 10.250.15.184 10.250.31.255 UDP 86 57621 → 57621 Len=44
180285 161.876329100 10.250.25.238 224.0.0.251 MDNS 295 Standard query response 0x0000 PTR SAICHARANREDDY._dosvc._tcp.local SRV 0 0 7680 SAICHARANREDDY.local TXT A 10.250.25.238 AAAA f
180286 161.876330300 fe80::6646:a12:ad1e… ff02::fb MDNS 315 Standard query response 0x0000 PTR SAICHARANREDDY._dosvc._tcp.local SRV 0 0 7680 SAICHARANREDDY.local TXT A 10.250.25.238 AAAA f
180287 161.876330900 HonHaiPrecis_43:b5:… Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.12.179
180288 161.876331900 fe80::7d00:33a9:932… ff02::fb MDNS 319 Standard query response 0x0000 PTR DESKTOP-L6B3HV6._dosvc._tcp.local SRV 0 0 7680 DESKTOP-L6B3HV6.local TXT A 10.250.19.87 AAAA
180289 161.876332500 10.250.19.87 224.0.0.251 MDNS 299 Standard query response 0x0000 PTR DESKTOP-L6B3HV6._dosvc._tcp.local SRV 0 0 7680 DESKTOP-L6B3HV6.local TXT A 10.250.19.87 AAAA
180290 161.876333100 Intel_0d:7b:e5 Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.10.219
180291 161.876334400 10.250.13.4 224.0.0.251 MDNS 617 Standard query response 0x0000 PTR Siddhartha._dosvc._tcp.local SRV 0 0 7680 Siddhartha.local TXT A 10.250.13.4 NSEC Siddhartha.
180292 161.876335500 fe80::b44c:9391:691… ff02::fb MDNS 473 Standard query response 0x0000 PTR DESKTOP-I8C8AJG._dosvc._tcp.local SRV 0 0 7680 DESKTOP-I8C8AJG.local TXT A 10.250.27.102 AAAA
180293 161.876336800 10.250.27.102 224.0.0.251 MDNS 453 Standard query response 0x0000 PTR DESKTOP-I8C8AJG._dosvc._tcp.local SRV 0 0 7680 DESKTOP-I8C8AJG.local TXT A 10.250.27.102 AAAA
180294 161.876346600 10.250.18.136 224.0.0.251 MDNS 629 Standard query response 0x0000 PTR DESKTOP-2L0MACQ._dosvc._tcp.local SRV 0 0 7680 DESKTOP-2L0MACQ.local TXT A 10.250.18.136 AAAA
180295 161.876349500 fe80::9e11:4d20:d79… ff02::fb MDNS 649 Standard query response 0x0000 PTR DESKTOP-2L0MACQ._dosvc._tcp.local SRV 0 0 7680 DESKTOP-2L0MACQ.local TXT A 10.250.18.136 AAAA
180296 161.876350100 CloudNetwork_7f:bd:… Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.23.127
180297 161.876351200 fe80::d31c:beb7:25d… ff02::fb MDNS 97 Standard query 0x0000 PTR _dosvc._tcp.local, "QU" question
180298 161.876351700 Intel_cf:55:76 Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.20.72
180299 161.876352400 WNC_89:69:fa Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.16.78
180300 161.876352900 Intel_78:ab:9f Broadcast ARP 56 Who has 10.250.24.234? Tell 10.250.11.195
180301 161.876353500 LiteonTechno_d1:11:… Broadcast ARP 56 Who has 10.250.17.60? Tell 10.250.5.225
180302 161.876354100 fe80::6f57:344d:ca5… ff02::1:ff5b:23fc ICMPv6 86 Neighbor Solicitation for fe80::9eb2:492c:655b:23fc from c4:ff:99:78:ab:9f
180303 161.879280700 fe80::694a:f16:67ed… ff02::1:ffd9:a047 ICMPv6 86 Neighbor Solicitation for fe80::d31c:beb7:25d9:a047 from 9c:67:d6:e9:92:8d
Frame 1: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Device\NPF_
Ethernet II, Src: Intel_6a:7f:2a (e8:b0:c5:6a:7f:2a), Dst: IPv4mcast_4d:4d:4d (01:00:5e:4d:4d:4d)
Internet Protocol Version 4, Src: 10.250.21.77, Dst: 224.77.77.77
User Datagram Protocol, Src Port: 63257, Dst Port: 12177
Data (113 bytes)
wireshark_Wi-FiBTQWY3.pcapng
Packets: 180303 · Dropped: 0 (0.0%)
Profile: Default

Step:4

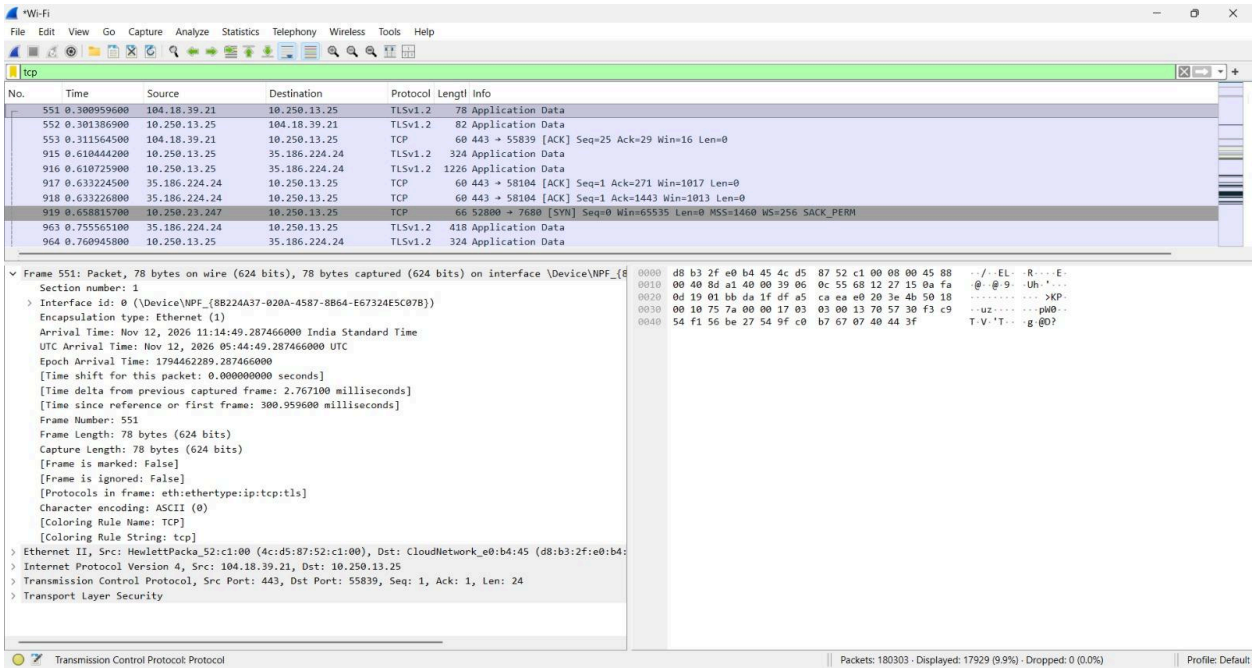

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
551 0.300959600 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
552 0.301386900 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
553 0.311564500 104.18.39.21 10.250.13.25 TCP 60 443 → 55839 [ACK] Seq=25 Ack=29 Win=16 Len=0
915 0.610444200 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
916 0.610725900 10.250.13.25 35.186.224.24 TLSv1.2 1226 Application Data
917 0.633224500 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=271 Win=1017 Len=0
918 0.633226800 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=1443 Win=1013 Len=0
919 0.658815700 10.250.23.247 10.250.13.25 TCP 66 52800 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
963 0.755565100 35.186.224.24 10.250.13.25 TLSv1.2 418 Application Data
964 0.760945800 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
965 0.761214300 10.250.13.25 35.186.224.24 TCP 1466 58104 → 443 [ACK] Seq=1713 Ack=365 Win=251 Len=1412 [TCP PDU reassembled in 966]
966 0.761214300 10.250.13.25 35.186.224.24 TLSv1.2 1274 Application Data
967 0.786756900 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=365 Ack=1713 Win=1012 Len=0
968 0.786758800 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=365 Ack=3125 Win=1007 Len=0
969 0.786760100 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=365 Ack=4345 Win=1003 Len=0
1381 0.923626000 35.186.224.24 10.250.13.25 TLSv1.2 393 Application Data
1382 0.961517400 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
1383 0.962998200 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
1384 0.967119500 104.18.39.21 10.250.13.25 TCP 60 443 → 56511 [ACK] Seq=25 Ack=29 Win=16 Len=0
1385 0.970639100 10.250.13.25 35.186.224.24 TCP 54 58104 → 443 [ACK] Seq=4345 Ack=704 Win=255 Len=0
1746 1.241484900 10.250.13.25 172.217.112.4 TCP 66 59150 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1835 1.355984300 149.154.170.100 10.250.13.25 TLSv1.2 190 Application Data
1836 1.361945800 172.217.112.4 10.250.13.25 TCP 66 443 → 59150 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1837 1.362112000 10.250.13.25 172.217.112.4 TCP 54 59150 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
1838 1.363564200 10.250.13.25 172.217.112.4 TCP 1466 59150 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 1839]
1839 1.363564200 10.250.13.25 172.217.112.4 TLSv1.3 434 Client Hello (SNI=www.googleapis.com)
1840 1.366292900 10.250.13.25 149.154.170.100 TLSv1.2 260 Application Data
1980 1.461766600 172.217.112.4 10.250.13.25 TLSv1.3 1466 Server Hello, Change Cipher Spec
1981 1.461769600 172.217.112.4 10.250.13.25 TCP 1466 443 → 59150 [PSH, ACK] Seq=1413 Ack=1793 Win=268032 Len=1412 [TCP PDU reassembled in 1984]
Frame 551: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{8
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:b4:
Internet Protocol Version 4, Src: 104.18.39.21, Dst: 10.250.13.25
Transmission Control Protocol, Src Port: 443, Dst Port: 55839, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
0000 d8 b3 2f e0 b4 45 4c d5 87 52 c1 00 08 00 45 88
0010 00 40 8d a1 40 00 39 06 0c 55 68 12 27 15 0a fa
0020 0d 19 01 bb da 1f df a5 ca ea e0 20 3e 4b 50 18
0030 00 10 75 7a 00 00 17 03 03 00 13 70 57 30 f3 c9
0040 54 f1 56 be 27 54 9f c0 b7 67 07 40 44 3f
Transmission Control Protocol: Protocol
Packets: 180303 · Displayed: 17929 (9.9%) · Dropped: 0 (0.0%)
Profile: Default

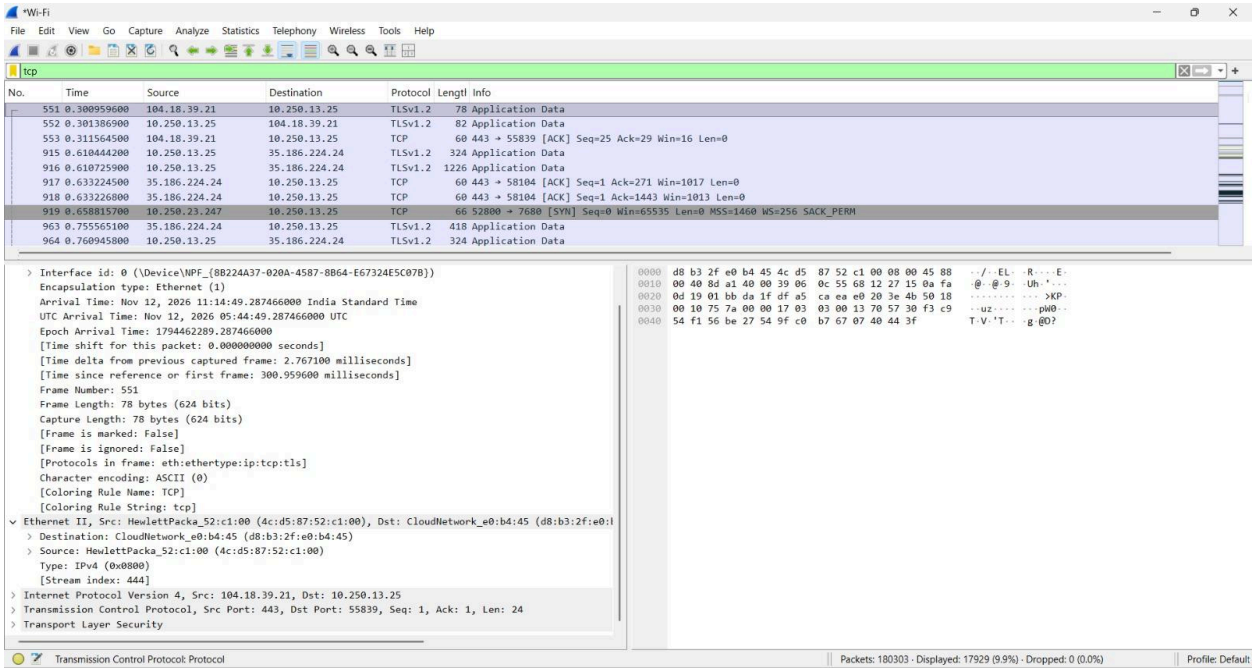

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp
udp
udpcp
udpencap
udplite
No. Source Destination Protocol Length Info
10.250.21.100 224.0.0.251 MDNS 319 Standard query response 0x0000 PTR Spotify Desktop Launcher(8)._spotify-connect._tcp.local SRV 0 0 62499 Srinivas.local TXT A 10
fe80::7f7d:324b:f92… ff02::fb MDNS 339 Standard query response 0x0000 PTR Spotify Desktop Launcher(8)._spotify-connect._tcp.local SRV 0 0 62499 Srinivas.local TXT A 10
10.250.25.62 10.250.31.255 NBNS 110 Registration NB DESKTOP-307I3LG<20>
536 0.295829900 0.0.0.0 255.255.255.255 DHCP 365 DHCP Request - Transaction ID 0x4622cdb6
537 0.295831800 10.250.24.110 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
538 0.295833100 fe80::50d1:1190:f95… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
539 0.295833800 10.250.24.110 224.0.0.251 MDNS 82 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
540 0.295834500 fe80::50d1:1190:f95… ff02::fb MDNS 102 Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
543 0.297200700 fe80::b44c:9391:691… ff02::fb MDNS 408 Standard query response 0x0000 PTR DESKTOP-I8C8AJG._dosvc._tcp.local SRV 0 0 7680 DESKTOP-I8C8AJG.local TXT
544 0.297202400 10.250.27.102 224.0.0.251 MDNS 388 Standard query response 0x0000 PTR DESKTOP-I8C8AJG._dosvc._tcp.local SRV 0 0 7680 DESKTOP-I8C8AJG.local TXT
545 0.297203100 10.250.27.102 224.0.0.251 MDNS 93 Standard query 0x0000 ANY DESKTOP-I8C8AJG._dosvc._tcp.local, "QM" question
546 0.297204000 fe80::b44c:9391:691… ff02::fb MDNS 113 Standard query 0x0000 ANY DESKTOP-I8C8AJG._dosvc._tcp.local, "QM" question
547 0.298180000 10.250.4.211 224.0.0.251 MDNS 470 Standard query response 0x0000 PTR Katakam's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 52
548 0.298190300 fe80::55:26de:6824:… ff02::fb MDNS 490 Standard query response 0x0000 PTR Katakam's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 52
549 0.298191500 10.250.8.173 239.255.255.250 SSDP 167 M-SEARCH * HTTP/1.1
554 0.381196200 fe80::363:4d2b:ae91… ff02::fb MDNS 341 Standard query response 0x0000 PTR Spotify Desktop Launcher(5)._spotify-connect._tcp.local SRV 0 0 62493 SRIANJANA.local TXT A 10
555 0.381198300 10.250.13.121 224.0.0.251 MDNS 321 Standard query response 0x0000 PTR Spotify Desktop Launcher(5)._spotify-connect._tcp.local SRV 0 0 62493 SRIANJANA.local TXT A 10
557 0.381200500 10.250.30.20 10.250.31.255 NBNS 110 Registration NB MAC-484462<00>
558 0.381201100 10.250.30.56 224.0.0.251 MDNS 321 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 63975 Praharshitha.local TXT A 10
559 0.382231700 fe80::d1e1:a2b8:c22… ff02::fb MDNS 331 Standard query response 0x0000 PTR Spotify Desktop Launcher(30)._spotify-connect._tcp.local SRV 0 0 53791 MSI.local TXT A 10.250
560 0.382849300 fe80::d81c:f2a4:881… ff02::fb MDNS 341 Standard query response 0x0000 PTR Spotify Desktop Launcher._spotify-connect._tcp.local SRV 0 0 63975 Praharshitha.local TXT A 10
561 0.382850400 10.250.21.47 224.0.0.251 MDNS 311 Standard query response 0x0000 PTR Spotify Desktop Launcher(30)._spotify-connect._tcp.local SRV 0 0 53791 MSI.local TXT A 10.250
565 0.382857300 10.250.2.166 224.0.0.251 MDNS 915 Standard query response 0x0000 PTR Manideep's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 70
566 0.384322000 fe80::fd:16f5:6c0f:… ff02::fb MDNS 935 Standard query response 0x0000 PTR Manideep's MacBook Air._companion-link._tcp.local TXT TXT, cache flush SRV, cache flush 0 0 70
567 0.384323400 10.250.17.11 224.0.0.252 LLMNR 67 Standard query 0xc9e3 ANY kornypc
568 0.384325900 10.250.25.35 239.255.255.250 UDP/XML 666 57354 → 3702 Len=624
569 0.384326900 10.250.25.154 224.0.0.251 MDNS 152 Standard query 0x0006 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast.
570 0.384327800 fe80::a86e:13ff:fe3… ff02::fb MDNS 172 Standard query 0x0006 PTR _googlecast._tcp.local, "QM" question PTR _%9E5E7C8F47989526C9BCD95D24084F6F0B27C5ED._sub._googlecast.
572 0.385953300 10.250.25.209 224.0.0.251 MDNS 345 Standard query response 0x0000 PTR Spotify Desktop Launcher(20). spotify-connect. tcp.local SRV 0 0 65388 Navadeep-Vitus.local T
Frame 549: Packet, 167 bytes on wire (1336 bits), 167 bytes captured (1336 bits) on interface \Device\NP
Ethernet II, Src: e2:5b:05:98:da:f5 (e2:5b:05:98:da:f5), Dst: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:fa)
Internet Protocol Version 4, Src: 10.250.8.173, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 38571, Dst Port: 1900
Simple Service Discovery Protocol
0000 01 00 5e 7f ff fa e2 5b 05 98 da f5 08 00 45 00
0010 00 99 c6 f9 40 00 ff 11 b0 b8 0a fa 08 ad ef ff
0020 ff fa 96 ab 07 6c 00 85 2f a2 4d 2d 53 45 41 52
0030 43 48 20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a 48
0040 4f 53 54 3a 20 32 33 39 2e 32 35 35 2e 32 35 35
0050 2e 32 35 30 3a 31 39 30 30 0d 0a 4d 41 4e 3a 20
0060 22 73 73 64 70 3a 64 69 73 63 6f 76 65 72 22 0d
0070 0a 4d 58 3a 20 31 0d 0a 53 54 3a 20 75 72 6e 3a
User Datagram Protocol: Protocol
Packets: 180303 · Displayed: 88757 (49.2%) · Dropped: 0 (0.0%)
Profile: Default

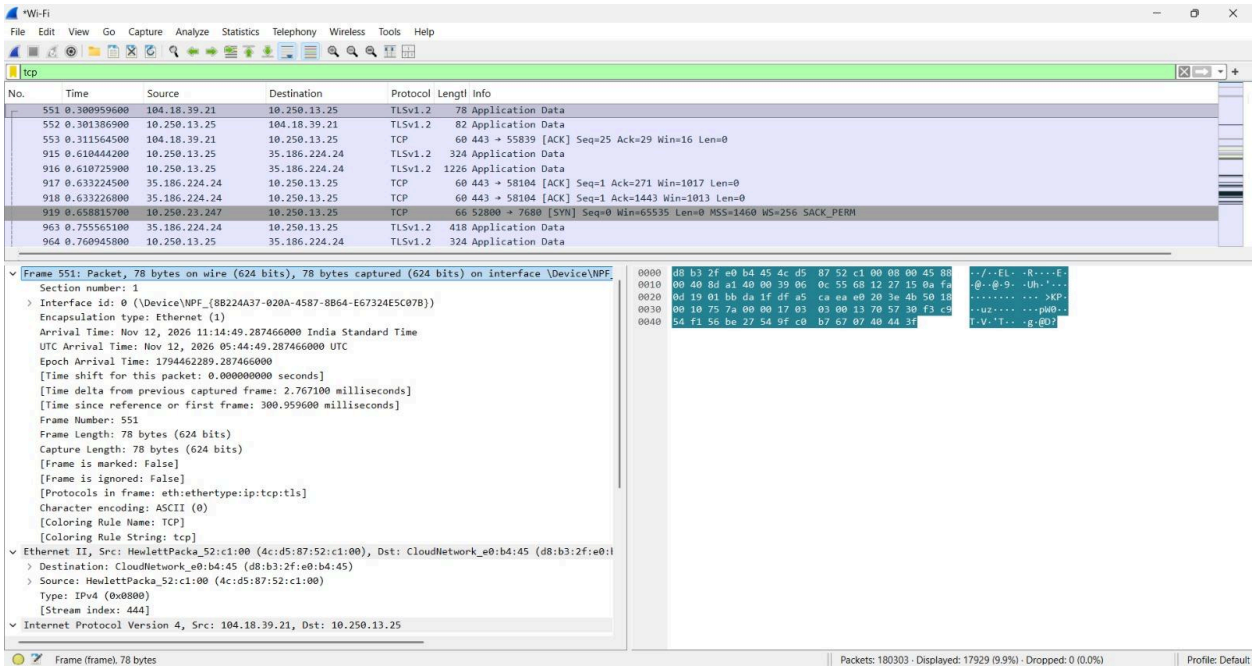

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Lengtl Info
Internet Control Message Protocol: Protocol
Packets: 180303 · Displayed: 0 (0.0%) · Dropped: 0 (0.0%)
Profile: Default

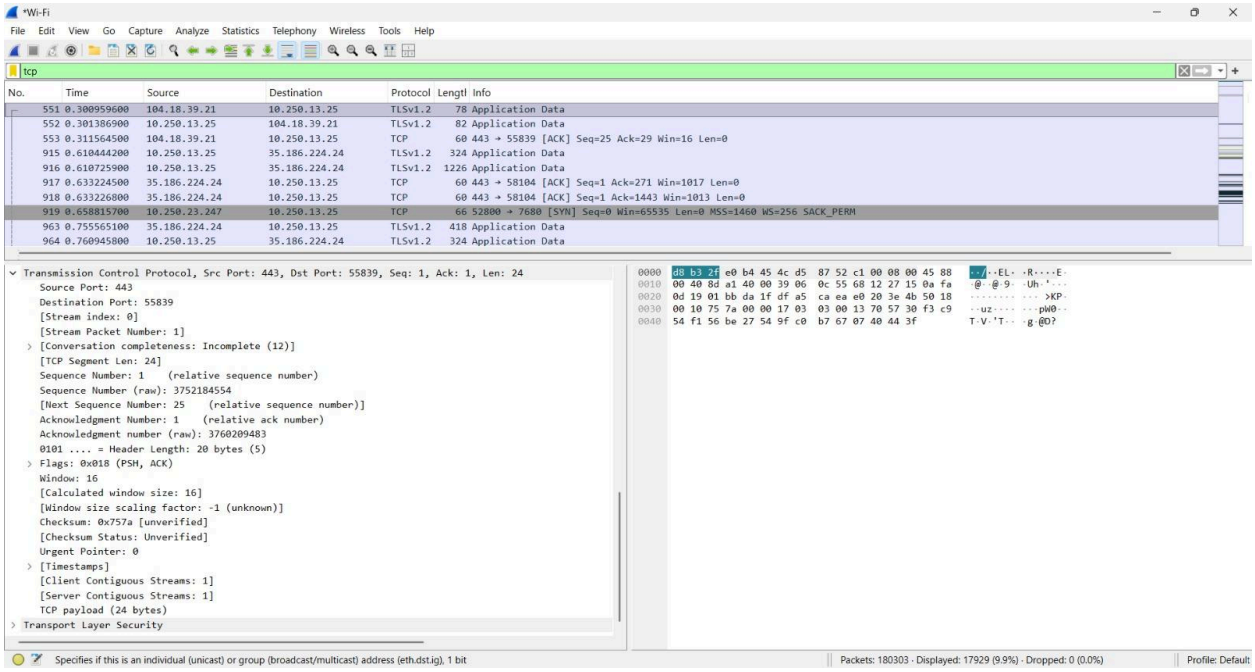

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.len > 0
No. Time Source Destination Protocol Lengtl Info
551 0.300959600 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
552 0.301386900 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
915 0.610444200 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
916 0.610725900 10.250.13.25 35.186.224.24 TLSv1.2 1226 Application Data
963 0.755565100 35.186.224.24 10.250.13.25 TLSv1.2 418 Application Data
964 0.760945800 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
965 0.761214300 10.250.13.25 35.186.224.24 TCP 1466 58104 → 443 [ACK] Seq=1713 Ack=365 Win=251 Len=1412 [TCP PDU reassembled in 966]
966 0.761214300 10.250.13.25 35.186.224.24 TLSv1.2 1274 Application Data
1381 0.923626000 35.186.224.24 10.250.13.25 TLSv1.2 393 Application Data
1382 0.961517400 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
1383 0.962998200 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
1835 1.355984300 149.154.170.100 10.250.13.25 TLSv1.2 190 Application Data
1838 1.363564200 10.250.13.25 172.217.112.4 TCP 1466 59150 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 1839]
1839 1.363564200 10.250.13.25 172.217.112.4 TLSv1.3 434 Client Hello (SNI=www.googleapis.com)
1840 1.366292900 10.250.13.25 149.154.170.100 TLSv1.2 260 Application Data
1980 1.461766600 172.217.112.4 10.250.13.25 TLSv1.3 1466 Server Hello, Change Cipher Spec
1981 1.461769600 172.217.112.4 10.250.13.25 TCP 1466 443 → 59150 [PSH, ACK] Seq=1413 Ack=1793 Win=268032 Len=1412 [TCP PDU reassembled in 1984]
1982 1.461771500 172.217.112.4 10.250.13.25 TCP 1466 443 → 59150 [ACK] Seq=2825 Ack=1793 Win=268032 Len=1412 [TCP PDU reassembled in 1984]
1984 1.463253000 172.217.112.4 10.250.13.25 TLSv1.3 243 Application Data
1986 1.466055700 10.250.13.25 172.217.112.4 TLSv1.3 144 Change Cipher Spec, Application Data
1987 1.466714700 10.250.13.25 172.217.112.4 TLSv1.3 146 Application Data
1988 1.467128200 10.250.13.25 172.217.112.4 TLSv1.3 323 Application Data
1989 1.467286500 10.250.13.25 172.217.112.4 TLSv1.3 87 Application Data
1990 1.482002700 10.250.13.25 149.154.167.99 TLSv1.2 244 Application Data
1991 1.499077000 172.217.112.4 10.250.13.25 TLSv1.3 1052 Application Data, Application Data
1992 1.499526700 10.250.13.25 172.217.112.4 TLSv1.3 85 Application Data
1993 1.499902000 172.217.112.4 10.250.13.25 TLSv1.3 85 Application Data
2123 1.562719700 172.217.112.4 10.250.13.25 TLSv1.3 330 Application Data
2124 1.562729300 172.217.112.4 10.250.13.25 TLSv1.3 358 Application Data
Frame 551: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{8
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:b4:
Internet Protocol Version 4, Src: 104.18.39.21, Dst: 10.250.13.25
Transmission Control Protocol, Src Port: 443, Dst Port: 55839, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
0000 d8 b3 2f e0 b4 45 4c d5 87 52 c1 00 08 00 45 88 ··/··EL· ·R····E·
0010 00 40 8d a1 40 00 39 06 0c 55 68 12 27 15 0a fa ·@··@·9· ·Uh·'···
0020 0d 19 01 bb da 1f df a5 ca ea e0 20 3e 4b 50 18 ········ ··· >KP·
0030 00 10 75 7a 00 00 17 03 03 00 13 70 57 30 f3 c9 ··uz···· ···pW0··
0040 54 f1 56 be 27 54 9f c0 b7 67 07 40 44 3f T·V·'T·· ·g·@D?
wireshark_Wi-FiBTQWY3.pcapng
Packets: 180303 · Displayed: 13205 (7.3%) · Dropped: 0 (0.0%)
Profile: Default

Step:5


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
551 0.300959600 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
552 0.301386900 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
553 0.311564500 104.18.39.21 10.250.13.25 TCP 60 443 → 55839 [ACK] Seq=25 Ack=29 Win=16 Len=0
915 0.610444200 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
916 0.610725900 10.250.13.25 35.186.224.24 TLSv1.2 1226 Application Data
917 0.633224500 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=271 Win=1017 Len=0
918 0.633226800 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=1443 Win=1013 Len=0
919 0.658815700 10.250.23.247 10.250.13.25 TCP 66 52800 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
963 0.755565100 35.186.224.24 10.250.13.25 TLSv1.2 418 Application Data
964 0.760945800 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
Frame 551: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{8
Section number: 1
Interface id: 0 (\Device\NPF_{8B224A37-020A-4587-8B64-E67324E5C07B})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 12, 2026 11:14:49.287466000 India Standard Time
UTC Arrival Time: Nov 12, 2026 05:44:49.287466000 UTC
Epoch Arrival Time: 1794462289.287466000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.767100 milliseconds]
[Time since reference or first frame: 300.959600 milliseconds]
Frame Number: 551
Frame Length: 78 bytes (624 bits)
Capture Length: 78 bytes (624 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:b4:
Internet Protocol Version 4, Src: 104.18.39.21, Dst: 10.250.13.25
Transmission Control Protocol, Src Port: 443, Dst Port: 55839, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
0000 d8 b3 2f e0 b4 45 4c d5 87 52 c1 00 08 00 45 88
0010 00 40 8d a1 40 00 39 06 0c 55 68 12 27 15 0a fa
0020 0d 19 01 bb da 1f df a5 ca ea e0 20 3e 4b 50 18
0030 00 10 75 7a 00 00 17 03 03 00 13 70 57 30 f3 c9
0040 54 f1 56 be 27 54 9f c0 b7 67 07 40 44 3f
Transmission Control Protocol: Protocol
Packets: 180303 · Displayed: 17929 (9.9%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
551 0.300959600 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
552 0.301386900 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
553 0.311564500 104.18.39.21 10.250.13.25 TCP 60 443 → 55839 [ACK] Seq=25 Ack=29 Win=16 Len=0
915 0.610444200 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
916 0.610725900 10.250.13.25 35.186.224.24 TLSv1.2 1226 Application Data
917 0.633224500 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=271 Win=1017 Len=0
918 0.633226800 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=1443 Win=1013 Len=0
919 0.658815700 10.250.23.247 10.250.13.25 TCP 66 52800 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
963 0.755565100 35.186.224.24 10.250.13.25 TLSv1.2 418 Application Data
964 0.760945800 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
Interface id: 0 (\Device\NPF_{8B224A37-020A-4587-8B64-E67324E5C07B})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 12, 2026 11:14:49.287466000 India Standard Time
UTC Arrival Time: Nov 12, 2026 05:44:49.287466000 UTC
Epoch Arrival Time: 1794462289.287466000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.767100 milliseconds]
[Time since reference or first frame: 300.959600 milliseconds]
Frame Number: 551
Frame Length: 78 bytes (624 bits)
Capture Length: 78 bytes (624 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:b4:
Destination: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:b4:45)
Source: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Type: IPv4 (0x0800)
[Stream index: 444]
Internet Protocol Version 4, Src: 104.18.39.21, Dst: 10.250.13.25
Transmission Control Protocol, Src Port: 443, Dst Port: 55839, Seq: 1, Ack: 1, Len: 24
Transport Layer Security
0000 d8 b3 2f e0 b4 45 4c d5 87 52 c1 00 08 00 45 88
0010 00 40 8d a1 40 00 39 06 0c 55 68 12 27 15 0a fa
0020 0d 19 01 bb da 1f df a5 ca ea e0 20 3e 4b 50 18
0030 00 10 75 7a 00 00 17 03 03 00 13 70 57 30 f3 c9
0040 54 f1 56 be 27 54 9f c0 b7 67 07 40 44 3f
Transmission Control Protocol: Protocol
Packets: 180303 · Displayed: 17929 (9.9%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
551 0.300959600 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
552 0.301386900 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
553 0.311564500 104.18.39.21 10.250.13.25 TCP 60 443 → 55839 [ACK] Seq=25 Ack=29 Win=16 Len=0
915 0.610444200 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
916 0.610725900 10.250.13.25 35.186.224.24 TLSv1.2 1226 Application Data
917 0.633224500 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=271 Win=1017 Len=0
918 0.633226800 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=1443 Win=1013 Len=0
919 0.658815700 10.250.23.247 10.250.13.25 TCP 66 52800 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
963 0.755565100 35.186.224.24 10.250.13.25 TLSv1.2 418 Application Data
964 0.760945800 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
Frame 551: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF
Section number: 1
Interface id: 0 (\Device\NPF_{8B224A37-020A-4587-8B64-E67324E5C07B})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 12, 2026 11:14:49.287466000 India Standard Time
UTC Arrival Time: Nov 12, 2026 05:44:49.287466000 UTC
Epoch Arrival Time: 1794462289.287466000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.767100 milliseconds]
[Time since reference or first frame: 300.959600 milliseconds]
Frame Number: 551
Frame Length: 78 bytes (624 bits)
Capture Length: 78 bytes (624 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:tls]
Character encoding: ASCII (0)
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00), Dst: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:
Destination: CloudNetwork_e0:b4:45 (d8:b3:2f:e0:b4:45)
Source: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Type: IPv4 (0x0800)
[Stream index: 444]
Internet Protocol Version 4, Src: 104.18.39.21, Dst: 10.250.13.25
0000 d8 b3 2f e0 b4 45 4c d5 87 52 c1 00 08 00 45 88
0010 00 40 8d a1 40 00 39 06 0c 55 68 12 27 15 0a fa
0020 0d 19 01 bb da 1f df a5 ca ea e0 20 3e 4b 50 18
0030 00 10 75 7a 00 00 17 03 03 00 13 70 57 30 f3 c9
0040 54 f1 56 be 27 54 9f c0 b7 67 07 40 44 3f
Frame (frame), 78 bytes
Packets: 180303 · Displayed: 17929 (9.9%) · Dropped: 0 (0.0%)
Profile: Default


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Lengtl Info
551 0.300959600 104.18.39.21 10.250.13.25 TLSv1.2 78 Application Data
552 0.301386900 10.250.13.25 104.18.39.21 TLSv1.2 82 Application Data
553 0.311564500 104.18.39.21 10.250.13.25 TCP 60 443 → 55839 [ACK] Seq=25 Ack=29 Win=16 Len=0
915 0.610444200 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
916 0.610725900 10.250.13.25 35.186.224.24 TLSv1.2 1226 Application Data
917 0.633224500 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=271 Win=1017 Len=0
918 0.633226800 35.186.224.24 10.250.13.25 TCP 60 443 → 58104 [ACK] Seq=1 Ack=1443 Win=1013 Len=0
919 0.658815700 10.250.23.247 10.250.13.25 TCP 66 52800 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
963 0.755565100 35.186.224.24 10.250.13.25 TLSv1.2 418 Application Data
964 0.760945800 10.250.13.25 35.186.224.24 TLSv1.2 324 Application Data
Transmission Control Protocol, Src Port: 443, Dst Port: 55839, Seq: 1, Ack: 1, Len: 24
Source Port: 443
Destination Port: 55839
[Stream index: 0]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (12)]
[TCP Segment Len: 24]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3752184554
[Next Sequence Number: 25 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3760209483
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 16
[Calculated window size: 16]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x757a [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (24 bytes)
Transport Layer Security
0000 d8 b3 2f e0 b4 45 4c d5 87 52 c1 00 08 00 45 88
0010 00 40 8d a1 40 00 39 06 0c 55 68 12 27 15 0a fa
0020 0d 19 01 bb da 1f df a5 ca ea e0 20 3e 4b 50 18
0030 00 10 75 7a 00 00 17 03 03 00 13 70 57 30 f3 c9
0040 54 f1 56 be 27 54 9f c0 b7 67 07 40 44 3f
Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit
Packets: 180303 · Displayed: 17929 (9.9%) · Dropped: 0 (0.0%)
Profile: Default

Step:6

Given:

- Ethernet Header = 14 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- Application Data = 24 bytes

1. Total Packet Size

= 14 + 20 + 20 + 24

= **78 bytes**

2. Total Header Overhead

= 14 + 20 + 20

= **54 bytes**

3. Payload Efficiency

= (Payload / Total Packet Size) × 100

= (24 / 78) × 100

= **30.77%**

4. Overhead Percentage

= (Header Overhead / Total Packet Size) × 100

= (54 / 78) × 100

= **69.23%**

Step:7

Parameter	TCP Packet	UDP Packet
Frame Length	78 bytes	167 bytes
Ethernet Header Length	14 bytes	14 bytes
IP Header Length	20 bytes	20 bytes
Transport Header Length	20 bytes	8 bytes
IP Total Length	64 bytes	153 bytes
Transport Header Length	20 bytes	8 bytes
Payload Length	24 bytes	125 bytes